Understanding Medications

Chapter 4: Enzymatic inhibitors Introduction

1

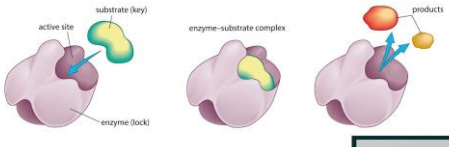
General ways in which our medications work (Pharmacodynamics)

* Direct action (rarely used)
* Receptor activity
  *Agonist
  *Antagonist
* Enzymatic inhibitor



2

Enzymatic inhibitor



Proton pump inhibitors inhibit the enzyme that secretes acid



3

Aims of this chapter

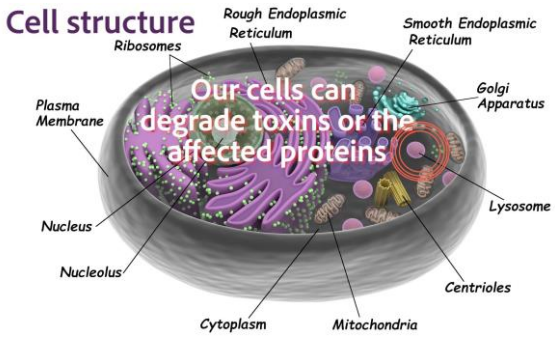
To apply the information about proton pump inhibitors to other medications

4

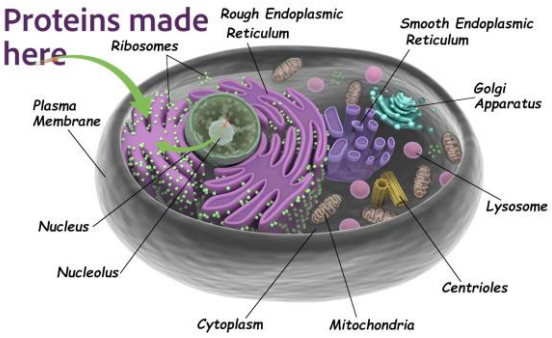
How long is an irreversible enzymatic inhibitor going to be effective?

a. They are 'irreversible' so they are permanent.
b. The effect is variable depending on enzyme and the tissue/cell involved.

5



6



7

Let's take this opportunity to review Pharmacokinetics before we get into the discussions of enzymes…

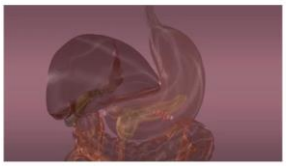
8

The Pharmacokinetics of Omeprazole and other proton pump inhibitors

Just picture the process
(Don't memorise the facts)

9

Omeprazole is enteric coated



(Does not dissolve in the stomach)

10

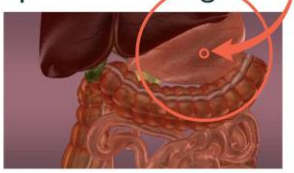
Omeprazole is enteric coated



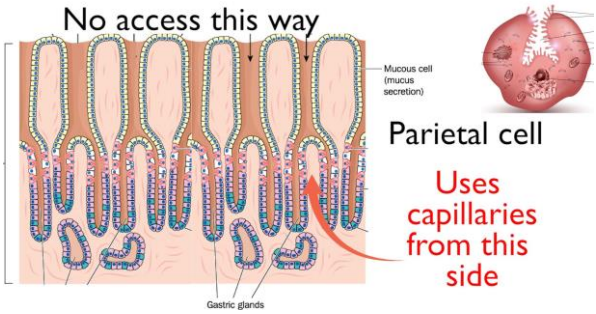
Absorbed in the small intestine

11

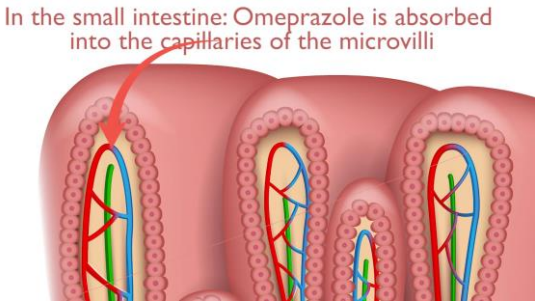
The cells that we affect with omeprazole are right here!



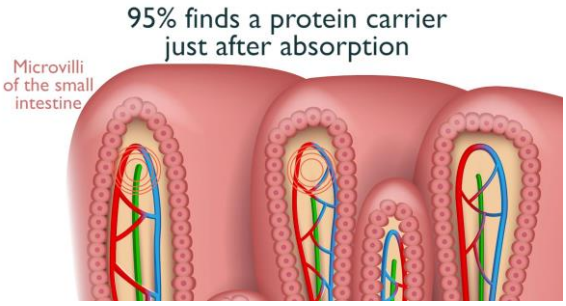
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No access this way
Mucous cell (mucus secretion)
Parietal cell
Uses capillaries from this side
Gastric glands

13


In the small intestine: Omeprazole is absorbed into the capillaries of the microvilli

14


95% finds a protein carrier just after absorption
Microvilli of the small intestine

15

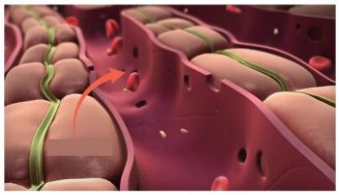
The liver has cytochrome P450 enzymes
CPY enzymes; CYP _ _ _



These metabolize drugs and toxins

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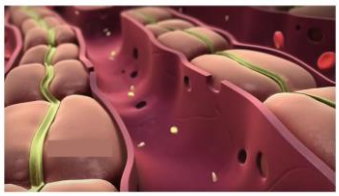
First pass: one enzyme alters the drug



The drug is released back into the blood

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Later: A phase II reaction will make the drug very water soluble



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Bioavailability

The fraction (%) of the administered dose that gets into systemic circulation unchanged

Intravenous administration (IV)	Oral administration (PO)
100%	Less than 100%

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The dose of the medication takes bioavailability into consideration

20

Why is Omeprazole is "well tolerated"?

A: The enzyme that is affected is only in:

* The parietal cell
* Some kidney cells (small amounts)

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Excretion of Omeprazole



80% is excreted by the kidneys

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Half life

Omeprazole has a half life of 1.2 - 1.5 hours

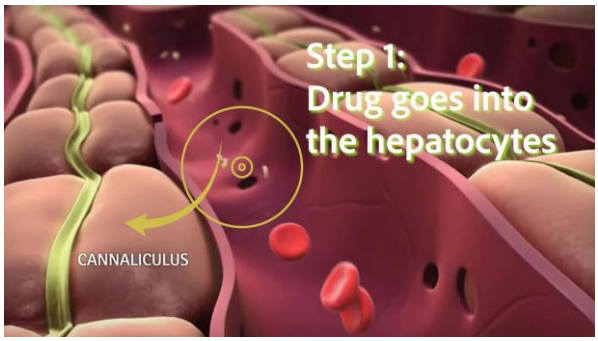
One would predict it needed to be taken frequently

...so why does it only need to be taken daily?

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Well done!
You can now understand the pharmacokinetics of any medication.

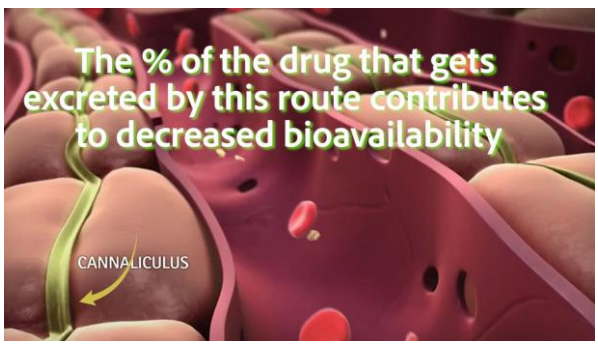
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Step 1:
Drug goes into
the hepatocytes
CANNALICULUS

25


BUT:
Some fat soluble
drugs go this way
CANNALICULUS

26


The % of the drug that gets
excreted by this route contributes
to decreased bioavailability
CANNALICULUS

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Omeprazole has a half-life of just 1 hour...

...but it is taken once daily

WHY?

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Different preparations of drugs

Enteric Coated Tablets
Capsules
Sustained Release Preparations
Controlled Release Tables
Oral Liquid Preparations
Topical Preparation
Sublingual and Buccal Administration.

Can release over time

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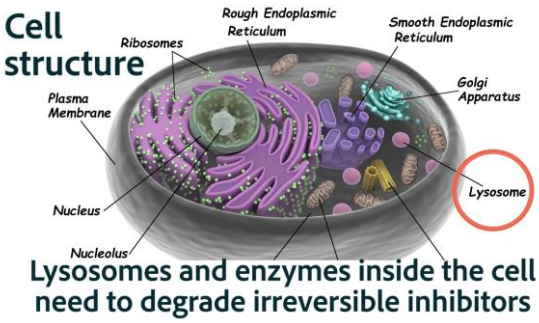
Different preparations of drugs

Enteric Coated Tablets
Capsules
Sustained Release Preparations
Controlled Release Tables
Oral Liquid Preparations
Topical Preparation
Sublingual and Buccal Administration.

X

Omepraxole is enteric coated, not released over time

30



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Congratulations, you can now apply your knowledge of Pharmacokinetics to any drug!

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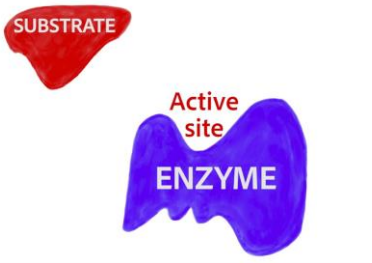
Enzymatic inhibition

What is an enzyme?

> A protein that catalyzes a reaction

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A normal enzymatic reaction



34

A normal enzymatic reaction





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Enzymatic inhibition

Reversible Irreversible

* Competitive
* Non-competitive
* Others

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Competitive inhibition

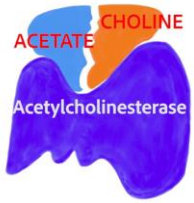
A medication that is a reversible enzymatic inhibitor



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Competitive Inhibition:

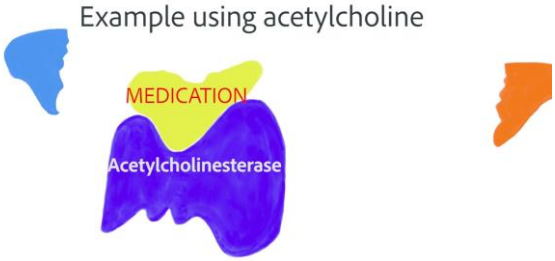
Example using acetylcholine



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Competitive Inhibition:

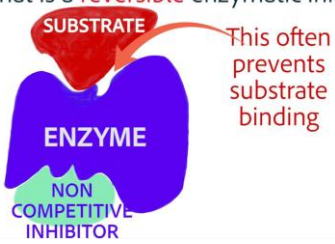
Example using acetylcholine



39

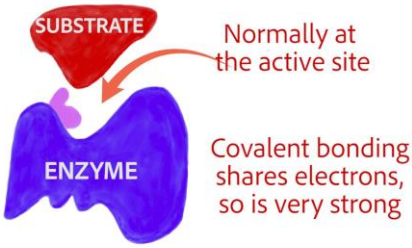
Non-competitive inhibition

A medication that is a reversible enzymatic inhibitor



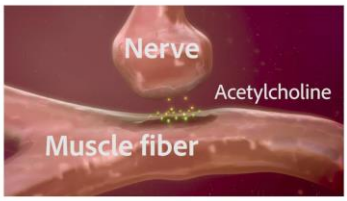
40

Irreversible enzymatic inhibitors



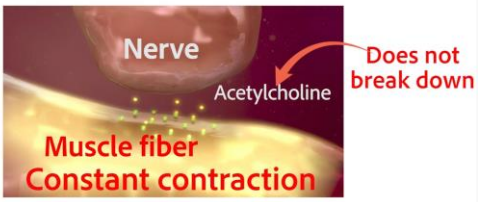
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Irreversible enzyme inhibitors: Nerve gas



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Irreversible enzyme inhibitors: Nerve gas



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Irreversible enzyme inhibitors: Penicillin

* Irreversibly inhibits the enzyme DD-transpeptidase
* Inhibit bacterial cell wall synthesis
* Humans do not have this enzyme



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Aspirin irreversibly inhibits COX

Platelets	Other tissues
For the entire 'life' of the platelet	For a few hours
Result:	**Result:**
Decreased platelet aggregation	Decreased inflammation

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Aspirin irreversibly inhibits COX

Platelets

For the entire life of the platelet

Result:

Decreased platelet aggregation

WHY?

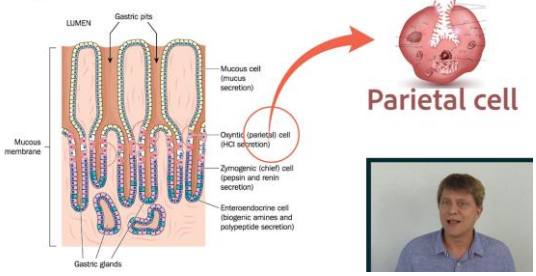


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Applying our knowledge of enzymes to the proton pump inhibitors

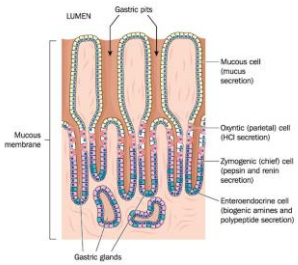
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Omeprazole affects a parietal cell enzyme



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Omeprazole affects a parietal cell enzyme

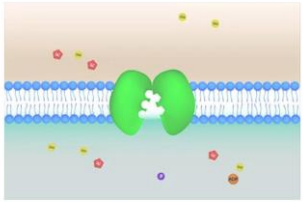




Histamine 2 antagonists block SOME of the stimuli for HCl release

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Proton pump inhibitors STOP THE PUMP



50

GOOD NEWS!

The hydrogen-potassium pump is only found in two locations

* The parietal cells
* The kidneys (not many)

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Which of the following is correct?
The H+/K+ pump is:

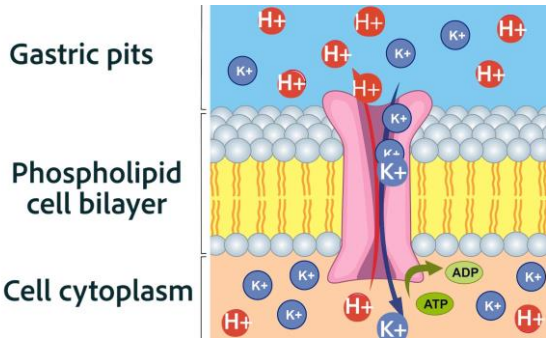
a. An enzyme because it catalyzes a reaction
b. A protein that is embedded in the parietal cells of the stomach
c. A protein that acts as a pump, pumping out hydrogen ions into the lumen of the stomach

52

Which of the following is correct?
The H+/K+ pump is:

a. An enzyme because it catalyzes a reaction
b. A protein that is embedded in the parietal cells of the stomach
c. A protein that acts as a pump, pumping out hydrogen ions into the lumen of the stomach
d. All of the above are correct

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Enzymatic inhibitors: Why do they have side effects?

Answer (first part): The enzyme may have other roles

Example: NSAIDs inhibit cyclooxygenase (COX)

* COX plays a major role in inflammation
* So NSAIDs decrease inflammation



55

Enzymatic inhibitors: Why do they have side effects?

Answer (first part): The enzyme may have other roles

Example:

Cyclooxygenase (COX) protects the Stomach and Kidneys



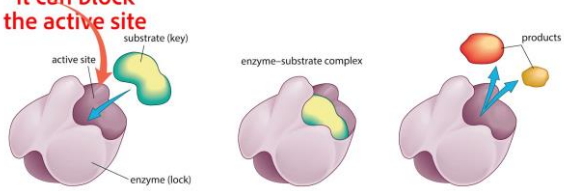




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Omeprazole has cross over effects on two liver enzymes

It can block the active site



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Expected interactions: Very variable

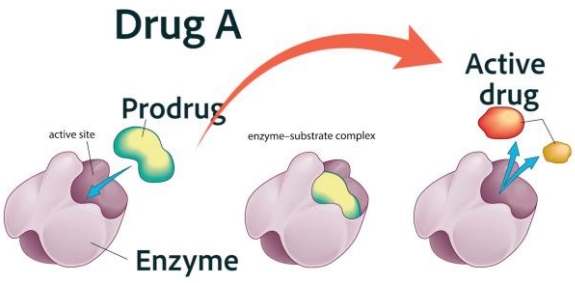
* Almost no effect
* Overdose of second medication
* Second medication metabolized quickly

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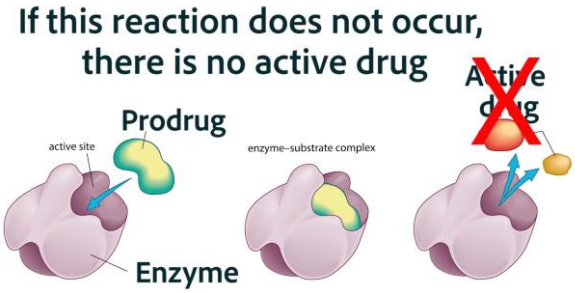
Grapefruit affects cytochrome P450 enzymes (CYP enzymes)

* Affects CYP3A4
* 50% of drugs are affected

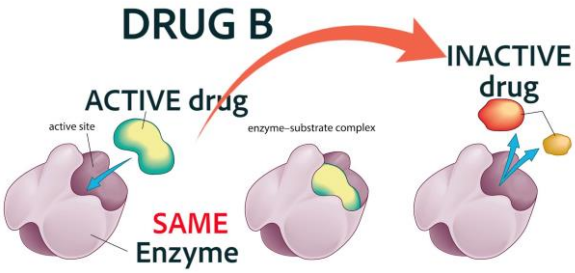
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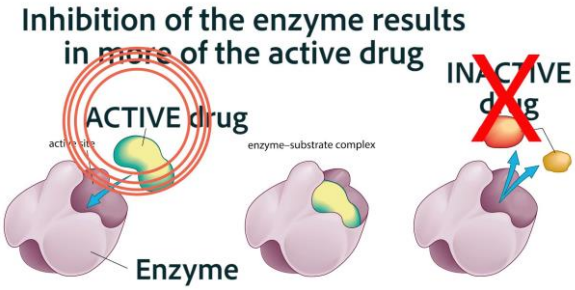
60


If this reaction does not occur, there is no active drug
Prodrug
active site
enzyme–substrate complex
Enzyme

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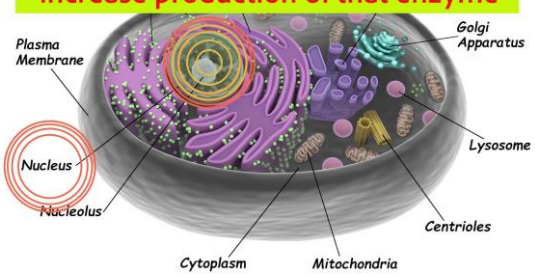

DRUG B
ACTIVE drug
INACTIVE drug
active site
enzyme–substrate complex
SAME
Enzyme

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Inhibition of the enzyme results in more of the active drug
ACTIVE drug
enzyme–substrate complex
Enzyme

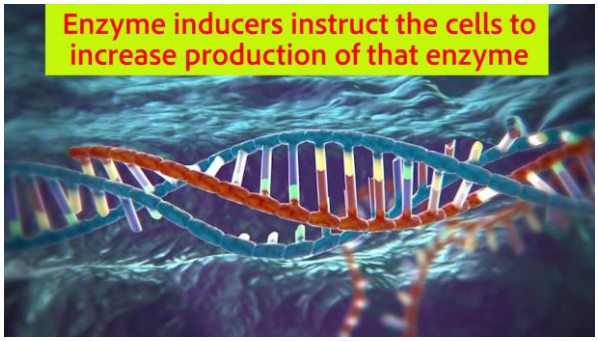
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Enzyme inducers instruct the cells to increase production of that enzyme



64

Enzyme inducers instruct the cells to increase production of that enzyme



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Omeprazole

Brand names: Prilosec, losec and others

Indications:

* GORD (GERD)
* Peptic ulcer disease
* Barrett's (o)esophagus



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Omeprazole: Indications

Barrett's oesophagus - a premalignant condition



Barrett's oesophagus can be caused by or worsened by ANY acid splashing on the oesophagus!

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Omeprazole: Indications

Peptic ulcer (can become a clinical emergency)



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Omeprazole:

Other side effects

* Increased risk of infections
  * GIT infections
  * Respiratory tract infections



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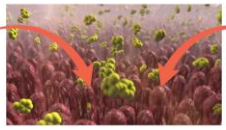
Omeprazole

Long term use may affect bone health

Why?

Absorption of some nutrients is decreased

Magnesium absorption decreased



Calcium absorption decreased

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Omeprazole

Inhibits CYP2C19 and CYP3A4

What?? These are cytochrome P450 enzymes that metabolize other medications

* That is your cue to look up medications that may interact with omeprazole

* If the client is not on other medications, there is no concern

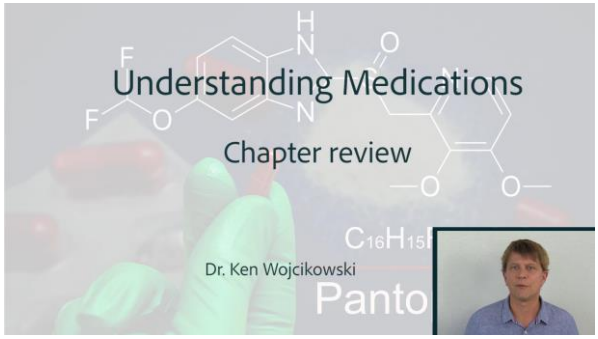
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Omeprazole

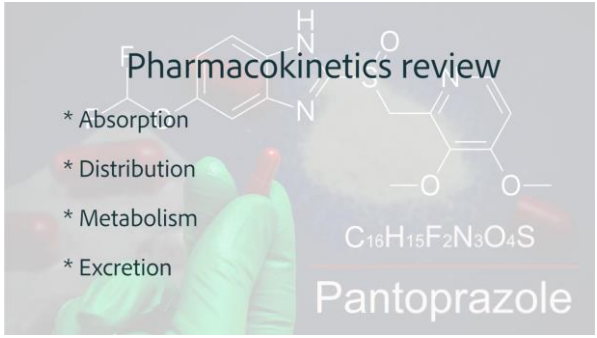
An irreversible inhibitor of the hydrogen-potassium ATPase, the enzyme that pumps acid

The effects of the medication will last much longer than the half-life would predict

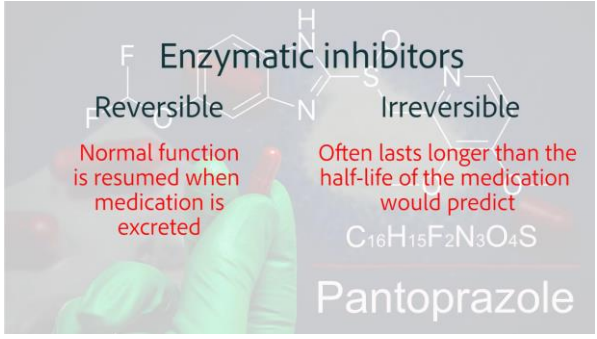
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Understanding Medications
Chapter review
Dr. Ken Wojcikowski
Panto

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Pharmacokinetics review
* Absorption
* Distribution
* Metabolism
* Excretion
C16H15F2N3O4S
Pantoprazole

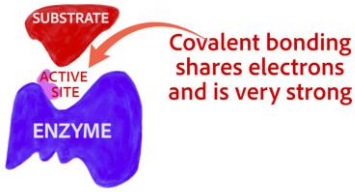
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Enzymatic inhibitors
Reversible
Irreversible
Normal function is resumed when medication is excreted
Often lasts longer than the half-life of the medication would predict
C16H15F2N3O4S
Pantoprazole

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Irreversible inhibitors

Medications that irreversibly inhibit an enzyme



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Omeprazole and other proton pump inhibitors are irreversible inhibitors

* Last much longer than the half-life would predict
* The enzyme affected is not vital for life

$C_{16}H_{15}F_2N_3O_4S$

Pantoprazole

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Pregnancy categories

(Australia and USA)

*A - Safest

* D - harmful to the fetus
* E - teratogenic (causes fetal abnormalities)

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Safety first

Never recommend withdrawing a medication without consulting the prescriber and getting more information

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Understanding Medications

Next: Neurotransmitters

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References

- Dr Ken Wojcikowski
- PharmacoPhoto
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